

Element I
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Importance to Tests:

We note the importance of each individual test:

1. Time
 - a. The time tracking must be accurate in order to shoot the food at the right time
2. Motors
 - a. The motor spinning must be smooth and uniform in order to spin the launch base properly
3. Launcher mech.
 - a. The launcher mechanism must shoot the food at a distance that gets it over the wall of moat

Testing Procedures:

Running the tests themselves turned out to be much less large scale than we wanted to. When it came to launching from the mech, the STEM building common area turned out to be the best possible area which gave us room to shoot. Regarding the code testing, it was much easier than the physical testing, and additionally, we made an algorithm for allocating the times for each day of the week rather than having to generate them manually each day, and we tested that algorithm in a third party IDE to see if it works.

Testing Results and Analysis

Time Code Testing

- The time code testing was very simple and thus seemed to have worked in getting the times of day without error. The only issue that could occur is rolling over on the hour as the code seems to have some issues at moving from 1:59 PM to 2:00 PM for example.

Algorithm Testing

- Due to being a late addition, we have not shown the code for the algorithm and thus it will be show below.

```

vector<int> randHourGen1 (map<int, int*>& passed_map, int day)
{
    //P - counting variable for upper limit hour
    //Z - Lower limit hour
    int p = 1;
    int z = 0;

    for(int i = 0; i < 1; i++)
    {
        int randomHour = static_cast<int>(random(passed_map[day][z],
passed_map[day][p]+1));
        returnVect.push_back(randomHour);
        int randomMinute = static_cast<int>(random(1, 60));
        returnVect.push_back(randomMinute);
        p = p + 2;
        z = z + 2;
    }
    return returnVect;
}

```

- Testing this code in a random IDE proved successful and in our 5 tests we were given 5 sets of two unique hours and two unique minutes

Motor Testing:

- In the testing of the motor with the servo, we noticed that the motors were moving at a speed that was a little too slow and we fear that the motor will not generate enough torque to move the launch mechanism
- Additionally, we do not have a clear idea of how to connect the motor to the arduino, and our motor code is still in development

Launching Mechanism

- In the testing of the launcher, we noticed that there was enough power to launch the food over the moat, according to the measured moats dimensions
- Additionally, the launcher seems to have a slight inherent randomness since the air pressure in the pneumatics system is non-ideal and thus the pressure changes which changes the launching distance

